

Meta

You are a Data Scientist on the WhatsApp consumer experience team focusing on enhancing user interaction with call and group chat features....

Question 1: How many distinct users have participated in group chats with names containing the word "family", where the chat was created in April 2024? This analysis will inform product managers about user engagement trends with family-focused chat groups.

```
SELECT COUNT(DISTINCT(user_id)) AS distinct_users
FROM fct_group_chats
WHERE chat_name ILIKE '%family%' AND (chat_creation_date BETWEEN '2024-04-01' AND '2024-04-30')
```



Meta

You are a Data Scientist on the WhatsApp consumer experience team focusing on enhancing user interaction with call and group chat features....

Question 2: To better understand user call behavior, we want to analyze the total call duration per user in May 2024. What is the average total call duration across all users?

```
SELECT AVG(total_call_duration) AS avg_total_call_duration
FROM (
    SELECT user_id, SUM(call_duration) AS total_call_duration
    FROM fct_user_calls
    WHERE call_date >= '2024-05-01' AND call_date < '2024-06-01'
    GROUP BY user_id
) AS user_call_totals;
```



Meta

You are a Data Scientist on the WhatsApp consumer experience team focusing on enhancing user interaction with call and group chat features....

Question 3: What is the maximum number of group chats any user has participated in during the second quarter of 2024 and how does this compare to the average participation rate? This insight will guide decisions on simplifying the chat interface for both heavy and average users.

```
with calculate AS (  
  SELECT user_id, COUNT(chat_id) AS num_group  
  FROM fct_group_chats  
  WHERE chat_creation_date >= '2024-04-01' AND chat_creation_date < '2024-07-01'  
  GROUP BY user_id)  
  
SELECT MAX(num_group) AS max_participation,  
       AVG(num_group) AS average_participation  
FROM calculate;
```

